SOUVIK KONAR

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SUMMARY

Final-year Computer Science and Engineering (Information Technology) undergraduate at SRM Institute of Science and Technology, graduating May 2027, with hands-on full-stack development experience across Python, Java, and JavaScript ecosystems, including React, Node.js/Express.js, and Django REST Framework. Skilled in designing and deploying REST APIs, relational database-backed applications, and cloud-hosted platforms (AWS Lambda, Render). Delivered a production-ready employee management platform (Django REST Framework, React, JWT auth) during a full-stack internship, alongside self-driven projects spanning real-time IoT systems, AI-assisted developer tooling, and a published IEEE research paper. Strong foundation in data structures, algorithms, and object-oriented design, with a track record of shipping reliable, well-structured software independently and under mentorship.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, C, C++, SQL
Frameworks & Libraries: React, Node.js, Express.js, Django REST Framework, Spring Boot, HTML/CSS
Core CS: Data Structures & Algorithms, Object-Oriented Design, DBMS, Operating Systems, REST API Design
Cloud & Databases: AWS Lambda, Render, MySQL, MongoDB, SQLite
AI/ML & Computer Vision: OpenAI API, PyTorch, TensorFlow, OpenCV, YOLOv8, Prompt Engineering
IoT, Protocols & APIs: ESP32, Arduino IDE, MQTT, REST, HTTP, WebSockets, JWT, Stripe API, Telegram Bot API
Tools & Practices: Git, GitHub, Agile Collaboration, Code Reviews, Problem Solving, Teamwork

INTERNSHIPS

Revolite Infotech Pvt. Ltd. — Python Full Stack Intern *Jun 2026 – Jul 2026*
Chennai, India | 45-day internship | Certificate

- Built Cadence, a full-stack employee management platform, as the primary project of the internship, using Django REST Framework and React
- Designed and developed RESTful APIs, relational database models, and JWT-based authentication under mentorship in a Python full-stack environment
- Recognized for sincere, dedicated, and result-oriented performance throughout the internship

Steel Authority of India Limited (SAIL) — Vocational Trainee *Dec 2025*
Durgapur Steel Plant, West Bengal | 11-day training | Certificate

- Completed vocational training across Blast Furnace, Coke Ovens, RMHP, Sinter Plant, and SMS departments
- Gained hands-on exposure to industrial manufacturing operations and plant-level engineering workflows

PROJECTS

Cadence — Employee Management Platform

- Built a full-stack employee management platform using Django REST Framework and React with JWT-based authentication and role-based access control
- Implemented attendance, leave, and payroll tracking modules backed by RESTful APIs and a relational database
- Deployed the application on Render with environment-based configuration for production readiness

RoadSense — Real-Time Pothole & Waste Detection System

- Developed an IoT-enabled monitoring system combining ESP32-CAM and Arduino edge devices with a custom-trained YOLOv8 model for dual-class pothole and waste detection
- Built a Flask-SocketIO backend for real-time inference and event streaming to a React and Leaflet geospatial dashboard
- Designed severity-based alerting logic and server-side input validation to filter low-confidence detections

IntelliCode — AI-Assisted Code Editor

- Built an AI-assisted code editor using React, Monaco Editor, and an Express.js backend integrated with the OpenAI API
- Designed REST endpoints for automated code debugging, explanation, and optimization with structured JSON responses
- Implemented request validation and configurable API routing for reliable client-server communication

RESEARCH PUBLICATION

Intelligent Home Automation System With Weather Forecasting and Alert Integration *Feb 2025 – Dec 2025*
13th IEEE International Conference | IEEE Xplore

- Developed NexHome, an IoT home automation system using ESP32 and multiple environmental sensors
- Implemented automation logic with Spring Boot and Telegram Bot API for real-time alerts
- Achieved a 30% improvement in energy efficiency through optimized IoT communication protocols

EDUCATION

B.Tech in Computer Science and Engineering with Information Technology *Expected May 2027*
SRM Institute of Science and Technology, Kattankulathur, Chennai | GPA: 8.83/10.00

CERTIFICATES & ACHIEVEMENTS

All certificates and achievements: Google Drive folder

- Intel Unnati AIoT Program (May 2025) — Trained in AIoT systems integrating AI and IoT
- SAP Generative AI Developer (Mar 2026 – Mar 2027) — Developed Generative AI applications using SAP BTP
- NPTEL Certifications: Programming in Java, Design and Analysis of Algorithms, Database Management Systems (2024–2025)
- Hackathons: Bitathon 2024 (Qualified 3/4 rounds), Smart India Hackathon 2024 (Internal Edition)